EXPERIMENT 9

Installation of MongoDB environment; Creation and Drop Database operations

Course Outcome	CO4	Teaching Hours	2

Objective

To install MongoDB Community Server and MongoDB Shell (mongosh)/Compass, and to perform basic database creation and deletion operations.

Theory

MongoDB is a document-oriented NoSQL database that stores data as flexible, JSON-like BSON documents grouped into collections (analogous to tables), rather than rigid rows/columns. mongosh is the modern MongoDB Shell for interacting with the database via commands; MongoDB Compass is the official GUI. In MongoDB, a database is not physically created until it contains at least one document — the use <dbname> command simply switches context to (or creates) a database, but it only persists once data is inserted into a collection.

Software / Hardware Requirements

MongoDB Community Server installer, MongoDB Compass, a Windows/Mac/Linux machine with admin rights.

Step-by-Step Procedure

1. Download the MongoDB Community Server installer for your OS from mongodb.com/try/download/community.
2. Run the installer, choose 'Complete' setup type, and ensure 'Install MongoDB as a Service' is checked so it starts automatically.
3. Also install MongoDB Compass (bundled with the installer, or download separately) for a graphical interface.
4. Verify installation by opening a terminal/command prompt and running mongod --version and mongosh --version.
5. Start the MongoDB service if not already running: on Windows check Services app for 'MongoDB Server'; on Mac/Linux run sudo systemctl start mongod (or brew services start mongodb-community).
6. Open a terminal and launch the shell by typing mongosh, which connects to the default local server at mongodb://localhost:27017.
7. List existing databases with the show dbs command to see the default admin, config and local databases.

```
test> show dbs
admin   40.00 KiB
config  60.00 KiB
local   40.00 KiB
```

8. Create/switch to a new database named collegeDB using the use command.

```
test> use collegeDB
switched to db collegeDB
```

9. Run show dbs again — notice collegeDB does NOT yet appear because it is empty (MongoDB creates the database lazily).

10. Insert one document into a collection to actually materialize the database.

```
collegeDB> db.students.insertOne({ name: "Parth Bhayani", rollNo: 512, branch: "Computer Engineering" })
{
  acknowledged: true,
  insertedId: ObjectId('6ababe0be36dab4ab645f9b0')
}
```

11. Run show dbs again and confirm collegeDB now appears in the list, since it contains data.

```
collegeDB> show dbs
admin       40.00 KiB
collegeDB   40.00 KiB
config      60.00 KiB
local       40.00 KiB
```

12. Verify the current database context with the db command, and list its collections with show collections.

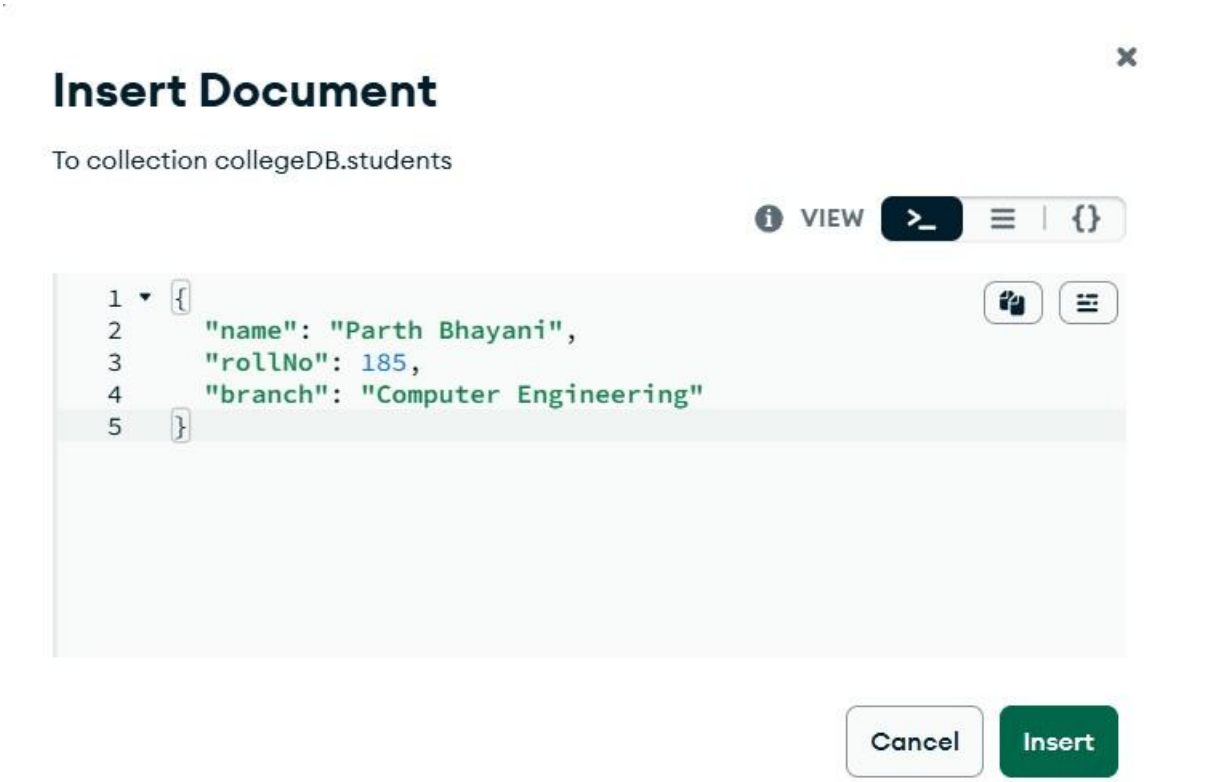
```
collegeDB> show dbs
admin       40.00 KiB
collegeDB   40.00 KiB
config      60.00 KiB
local       40.00 KiB
collegeDB> show collections
students
```

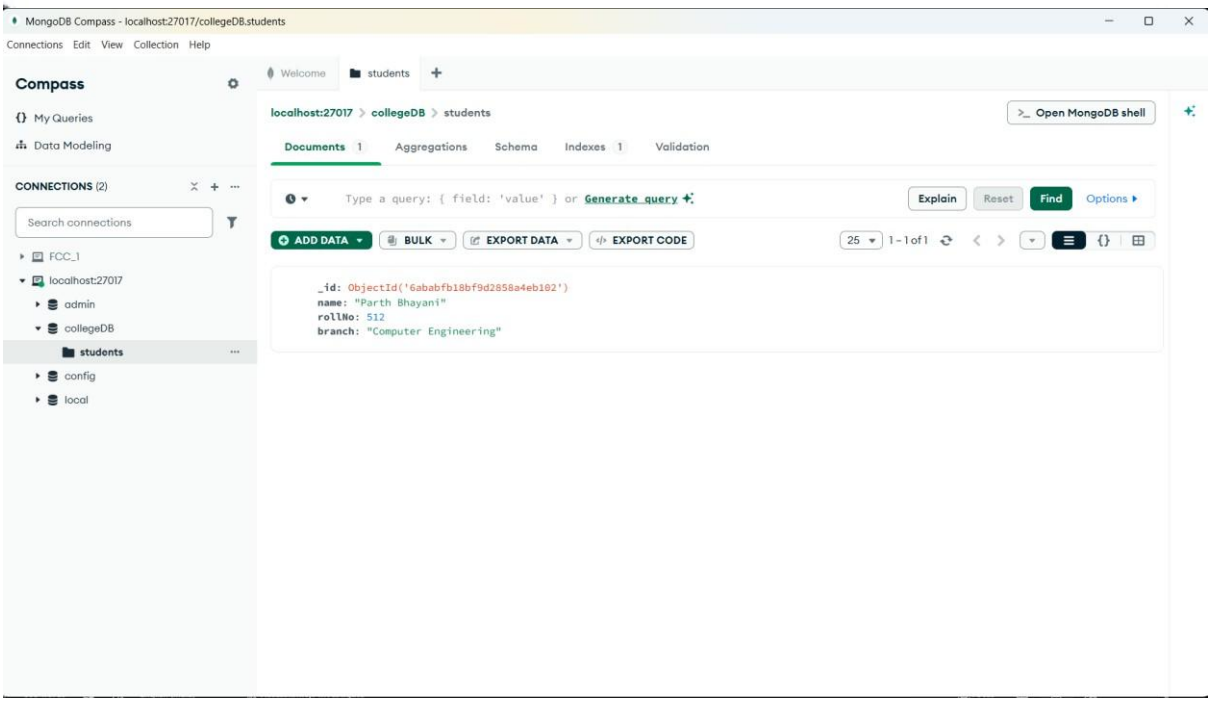
13. Drop the database entirely using db.dropDatabase() while collegeDB is the active context, then confirm with show dbs that it no longer appears.

```
collegeDB> db.dropDatabase()
{ ok: 1, dropped: 'collegeDB' }
collegeDB>
```

14. Open MongoDB Compass, connect to localhost:27017, and repeat the create/drop database exercise through the GUI to compare it with the shell workflow.

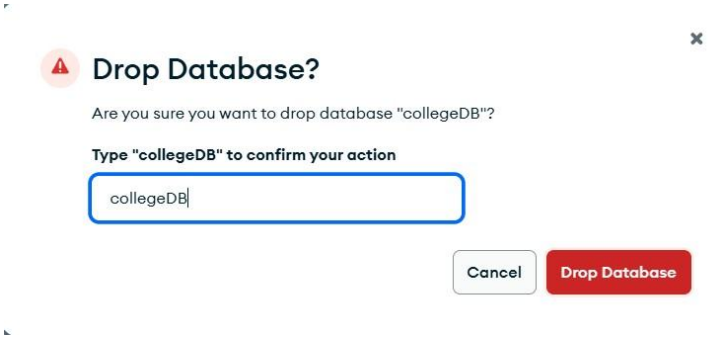
Name :- Parth Bhayani Prn No :- 26070122512 Batch:- B3
WAD





Delete Database :-

Name :- Parth Bhayani Prn No :- 26070122512 Batch:- B3
WAD



Volkswagen Database :-

```
test> use volkswagenDB
switched to db volkswagenDB
volkswagenDB> db.cars.insertOne({
|   model: "Virtus",
|   trim: "GT Plus",
|   engine: "1.5L TSI",
|   fuelType: "Petrol",
|   transmission: "7-Speed DSG",
|   year: 2024,
|   priceINR: 1940000,
|   inStock: true
| })
{
  acknowledged: true,
  insertedId: ObjectId('6abb372678fe4ff94581e7c3')
}
```

Healthcare Database :-

```
volkswagenDB> use healthcareDB
switched to db healthcareDB
healthcareDB> db.patients.insertOne({
|   patientId: "MED-2024-001",
|   name: "Arjun Verma",
|   age: 34,
|   gender: "Male",
|   bloodGroup: "O+",
|   department: "Cardiology",
|   attendingPhysician: "Dr. Nair",
|   admissionDate: new Date(),
|   status: "Admitted"
| })
{
  acknowledged: true,
  insertedId: ObjectId('6abb374578fe4ff94581e7c4')
}
```

Conclusion :- We have successfully downloaded and installed mongodb server on our systems .